



RoCam.

High Performance Vision-Guided Rocket Tracker

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Arch. Backend & Firmware

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Data Tooling & i18n

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CV, ML & System Tests

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TEAM #3 SPACEY | MCMASTER UNIVERSITY SFWRENG 4G06 | MCMASTER ROCKETRY TEAM



THE CHALLENGE

Rockets reach **Mach 3** in seconds.

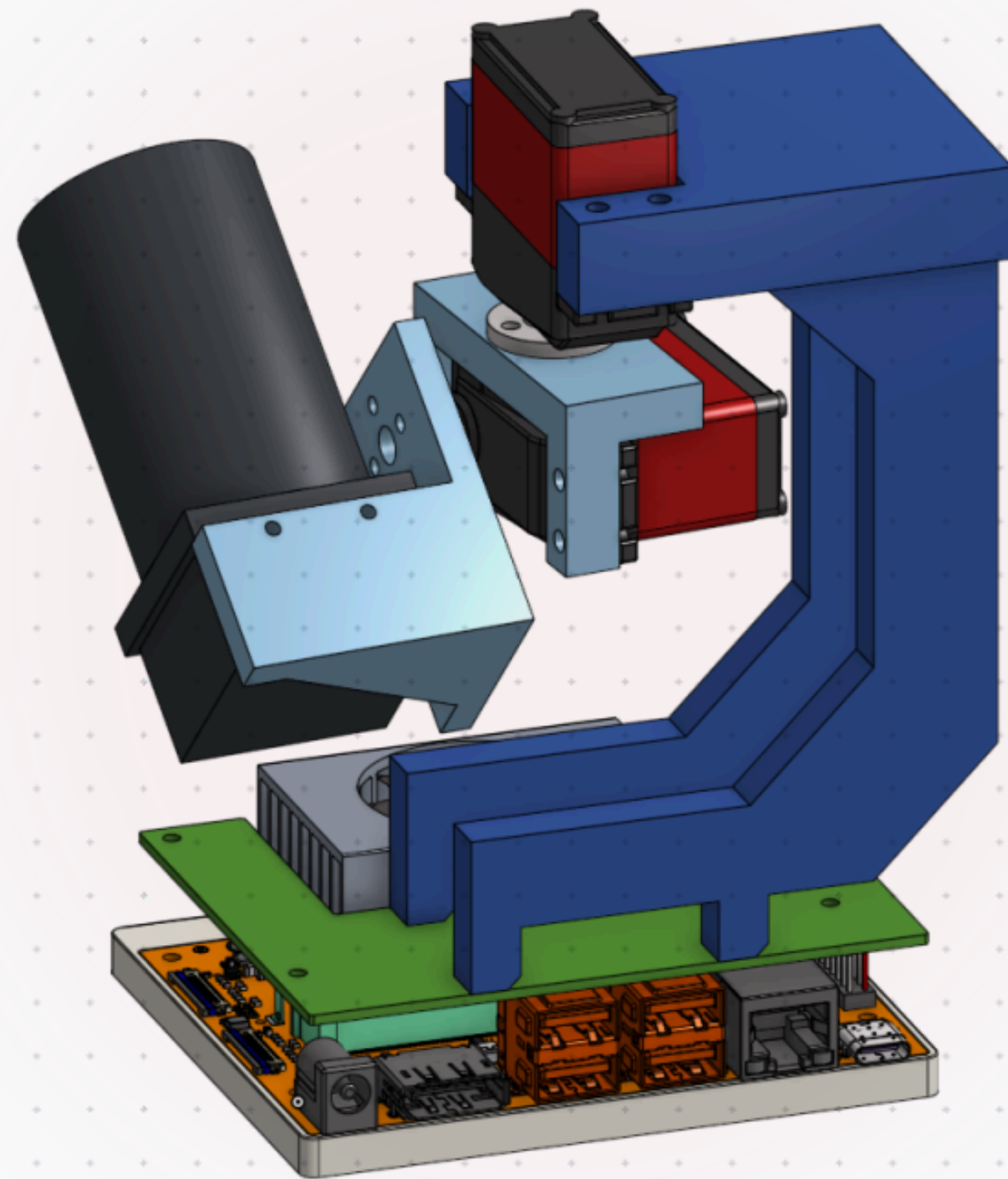
Manual optical tracking is physically impossible for small-profile hobbyist rocketry. Commercial rigs cost upwards of \$10,000.

100 km
MAX ALTITUDE

<1 s
EVENT DURATION

PRESENTATION ROADMAP

- 1. Problem & Solution
- 2. Architecture & Live Demo
- 3. Data, CV & ML Pipeline
- 4. Hardware & UI Control
- 5. Testing, Challenges & Future



Autonomous. Accessible. Accurate.

A fully autonomous, closed-loop optical tracking system engineered to meet strict capstone constraints.



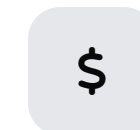
Edge AI Tracking

YOLO + TensorRT on Jetson Orin Nano



High-Speed Loop

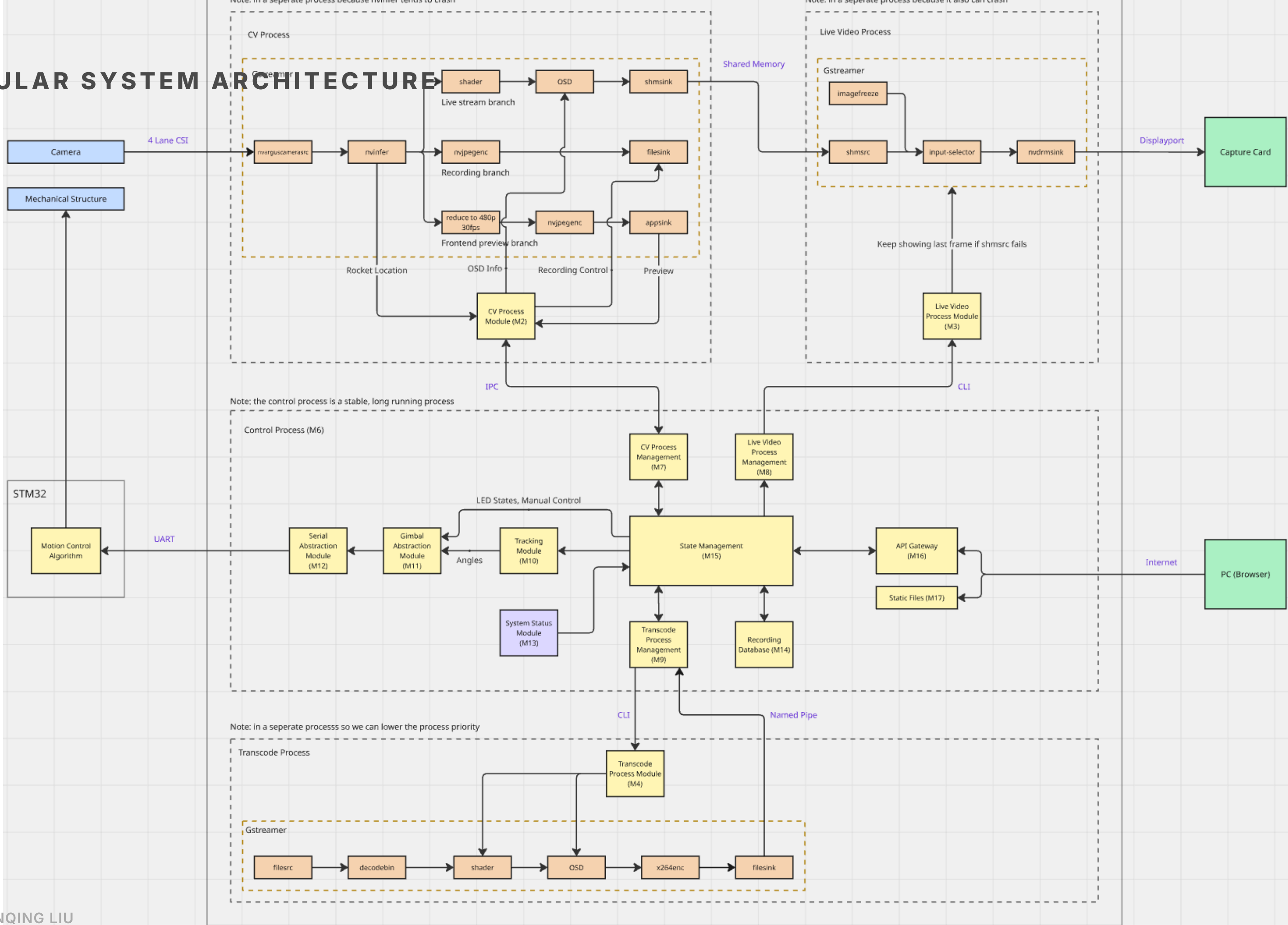
60Hz closed-loop servo actuation



Budget Constraint Met

Total BOM under \$500 CAD (vs \$10k+ commercial)

MODULAR SYSTEM ARCHITECTURE





LIVE DEMO

AUTONOMOUS TRACKING IN ACTION

Custom labelling platform for rocket data.

All training data flows through our in-house **RoCam Labeler** — a full-stack annotation tool built from scratch to handle our unique dataset requirements.



SAM-Assisted Annotation

One-click segmentation masks via Meta SAM



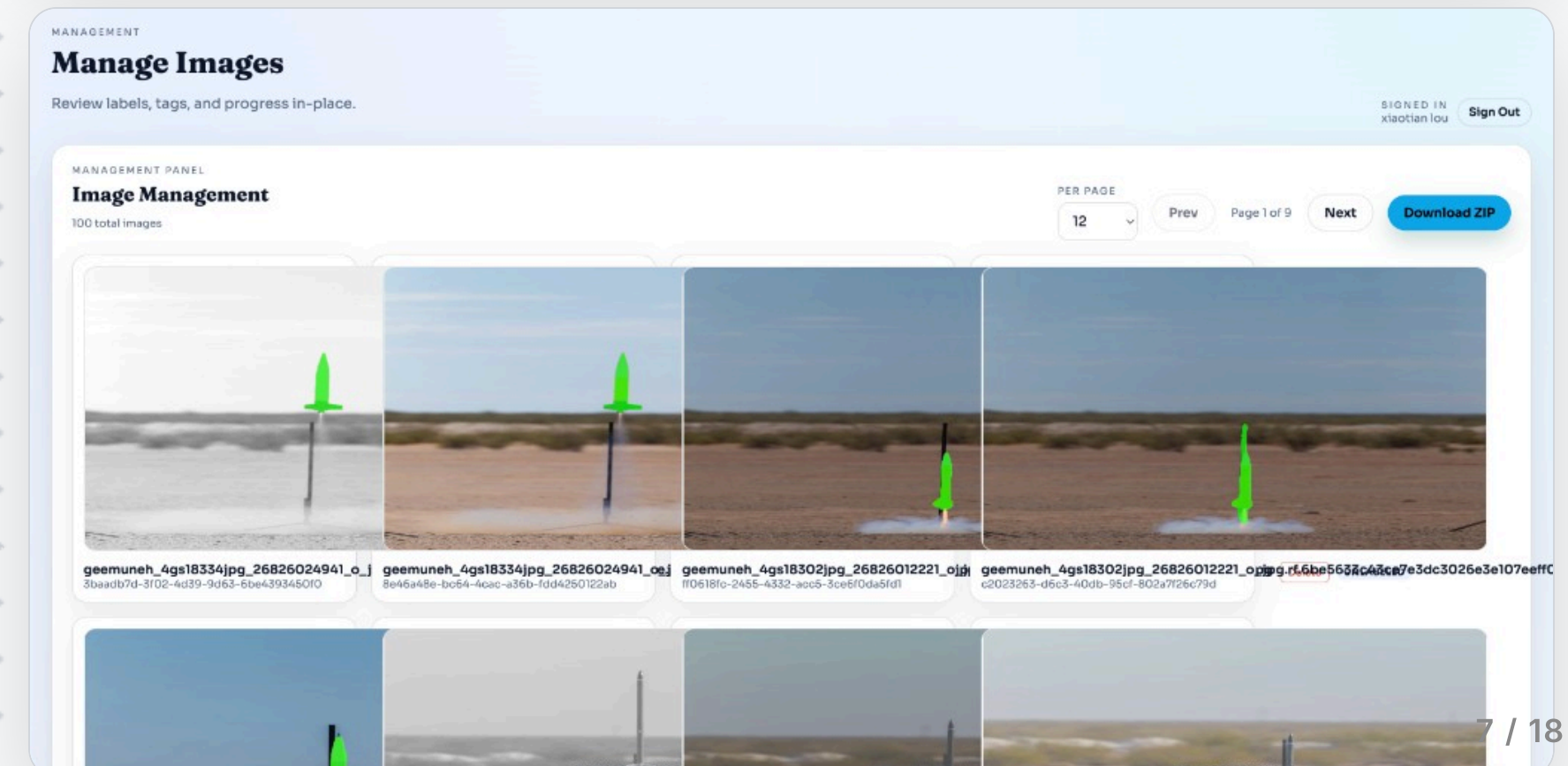
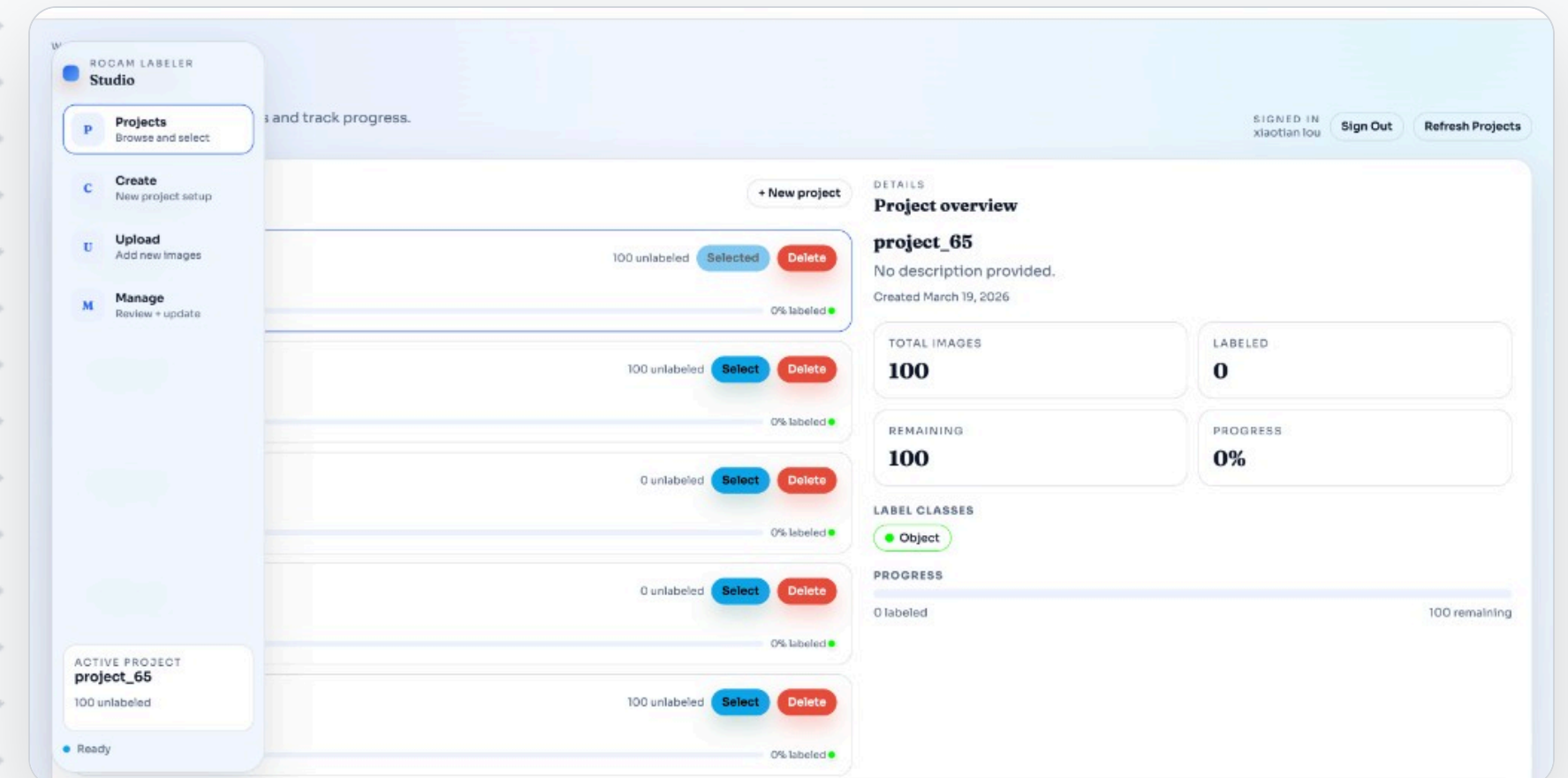
Project Management

Multi-project workspace with progress tracking

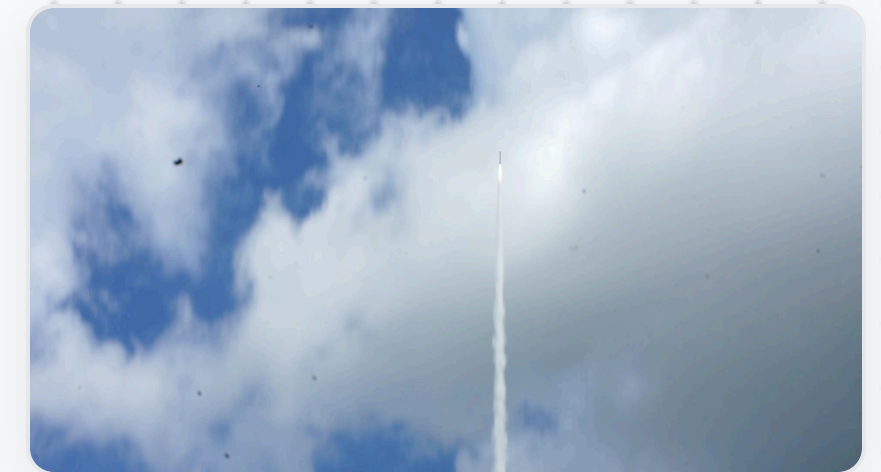


Batch Upload / Export

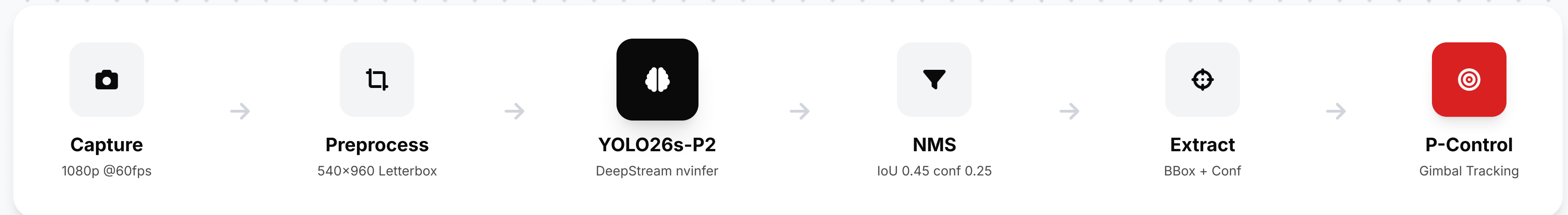
ZIP-based bulk import and YOLO-format export



Optimized for extreme distance and strict latency budgets.



Raw frame — can you spot the rocket?



🔍 Why YOLO26s-P2?

Custom P2 stride-4 detection head excels at micro-objects. Critical when the rocket occupies <1% of the frame at extreme range.

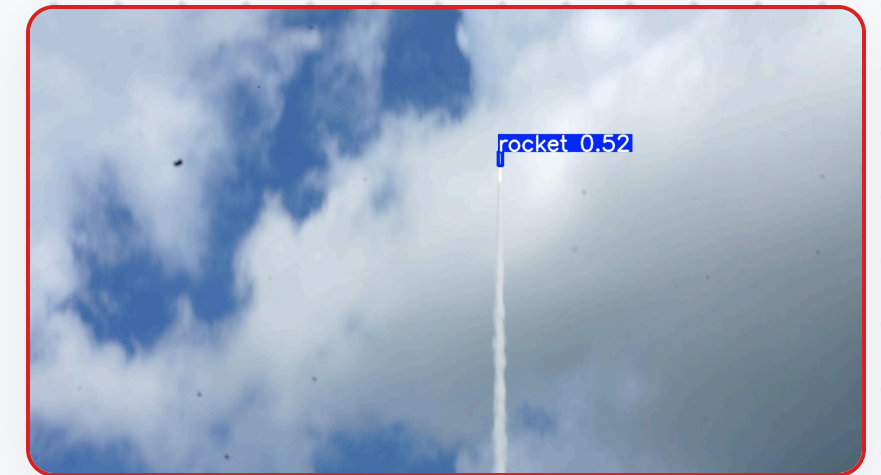
🔧 DeepStream Pipeline

DeepStream's nvinfer plugin handles the full pipeline on GPU — preprocessing, TensorRT FP16 inference, and NMS all under 16.6ms per frame at 60fps.

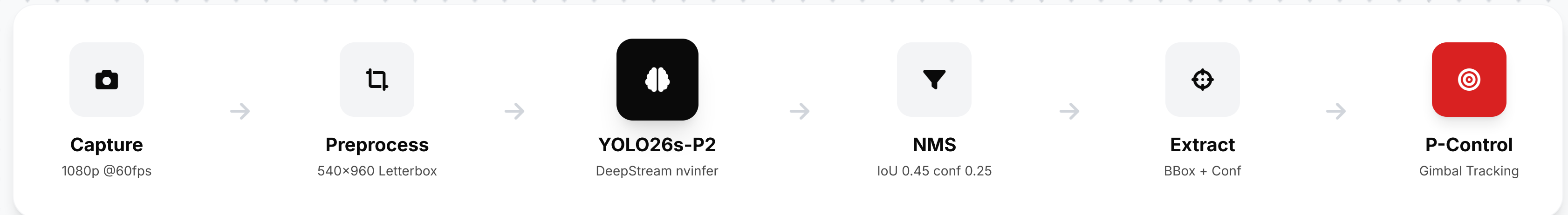
🎯 Gimbal Control

Proportional controller ($k_p=0.005$) converts pixel error to gimbal delta. IPC via Unix socket keeps E2E latency ~30ms.

Optimized for extreme distance and strict latency budgets.



Target <0.1% of frame — detected at conf 0.52



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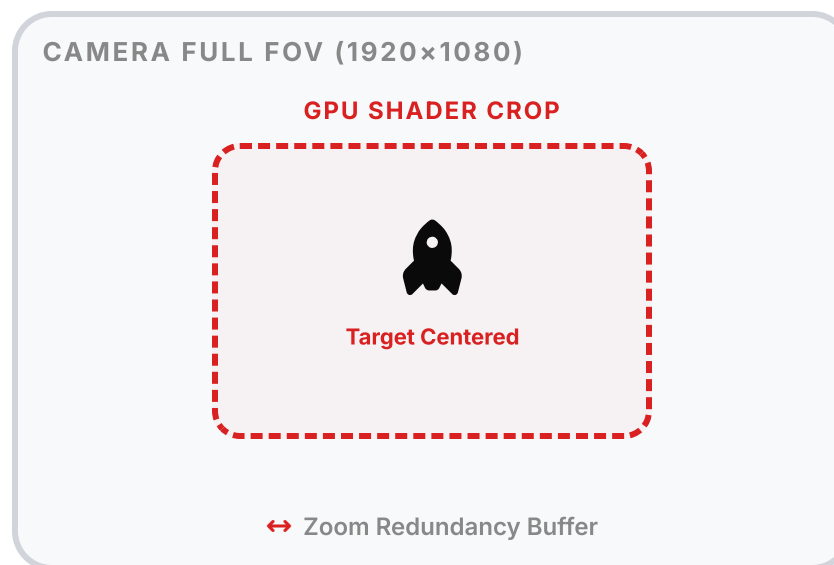
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Optical redundancy enables perceptually instant tracking.



FOV Redundancy Model

Wide optical view guarantees target stays in frame while gimbal mechanically catches up



Detection Result

BBox (cx, cy, w, h) from YOLO @ 60fps

~1MS



GPU Shader Path

- 1 Compute tx, ty, scale from bbox
- 2 GL shader crops & centers frame
- 3 Livestream output → operator sees instant lock

~100MS



Gimbal Servo Path

- 1 P-control → delta pan/tilt
- 2 UART → STM32 → PWM servo
- 3 Mechanical motion re-centers optical FOV



Zero Perceived Latency

Operator sees target locked within one display frame (~16ms). Gimbal catches up transparently.

0.958

FINAL MAP@50 SCORE

+22%

MAP@50-95 IMPROVEMENT

Over baseline COCO weights during extreme weather validation.

3-Stage Training Pipeline

Training data labelled via our custom **Labelling App** (built by Mike Chen) with SAM-assisted annotation.

1

Stage 1: Full Training (300 epochs)

COCO pre-trained YOLO26s-P2 fine-tuned on rocket dataset with mosaic augmentation (0.4) and zero patience.

2

Stage 2: Rect Fine-tune

Resume with rect=True and hard-negative backgrounds (empty skies, birds) to suppress false positives.

3

Stage 3: Low-LR Polish

Final epochs at reduced learning rate on curated, manually verified subset for maximum precision.

Bare-metal **Rust** for hard real-time execution.

60 Hz

CONTROL LOOP

Embassy

RUST ASYNC

CRC-8

DATA INTEGRITY

<0.5s

WATCHDOG TRIP



Jetson Orin

Vision / Math

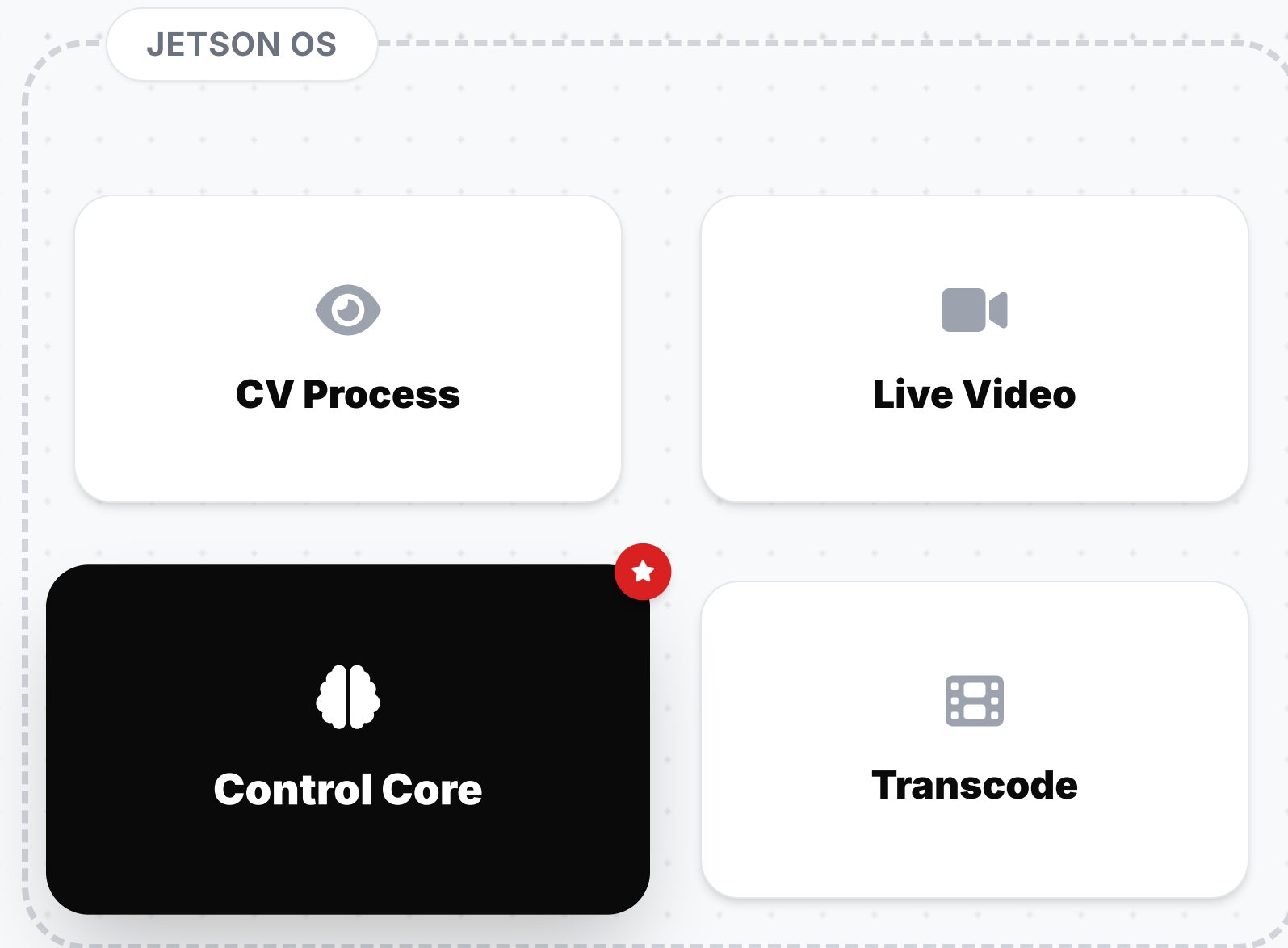
115200 BAUD UART

STM32F405

Custom PCB



FAULT-TOLERANT ARCHITECTURE



Built for isolation and real-time survival.



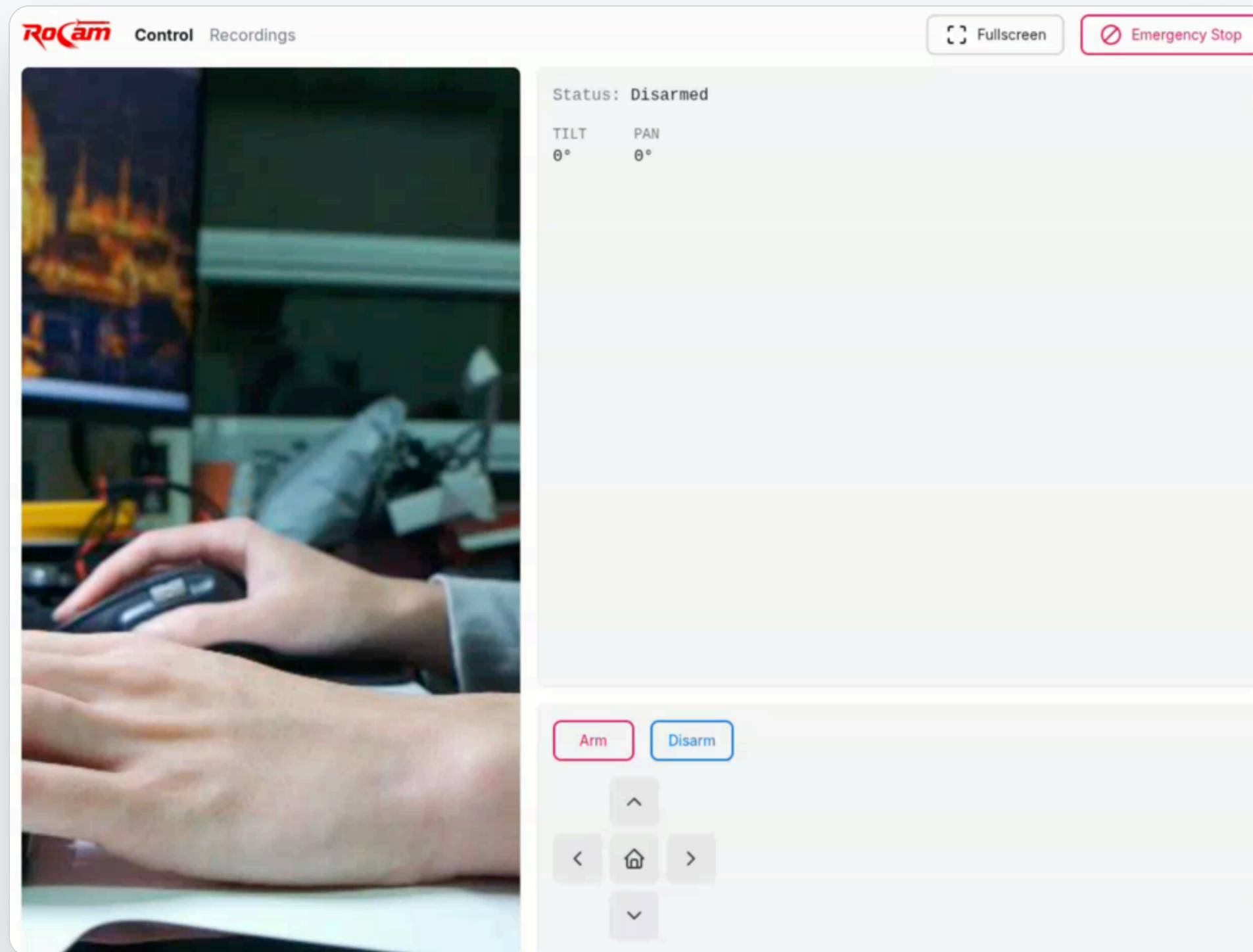
Independent Restart

If the CV inference crashes, the control process maintains connection to the gimbal, preventing lock-up.



Zero-copy IPC

Shared-memory architecture allows massive 1080p frames to move between processes instantly.



Command center in a browser.

REACT • TYPESCRIPT • VITE

- 🎥 Low-latency live preview
- 🎮 Direct manual gimbal override
- 📈 Real-time health telemetry
- 🌐 Full Bilingual (EN/FR) support

481

AUTOMATED TESTS PASSED



Backend (pytest)

313 Unit Tests

88% Coverage



Frontend (Vitest)

168 Component Tests

86% Coverage



System E2E

GitHub Actions CI

100% Pass Rate

PROJECT CHALLENGES

Real-time Latency

Python's GIL blocking CV loops.



Solution

Shared-memory IPC bypassing GIL entirely.

Micro-Object Detection

Rockets vanishing into background noise.



Solution

YOLO26s-P2 Head & Hard-negative mining.

Hardware Sync

UART dropping packets at 115200 baud.



Solution

Logic analyzer debug + custom CRC-8 checksums.

Designed to scale from 200m hobby rockets to 3km+ high-power launches.



Performance

- 4K @ 60fps tracking
- TensorRT INT8 Quantization
- Optical Auto-focus



Hardware Scaling

- Full-size Gimbal Mounts
- Multi-camera Arrays
- Weatherproof Enclosures

NEXT STEP



Software UX

- Mobile-native Apps
- Post-flight Analytics
- AR Telemetry Overlay



Thank You.

Questions?



github.com/ZifanSi/vision-guided-tracker

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